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Fire Services Learnings

Hazira gas terminal vapour release and fire, 24 September 2020. Draft learnings for ONGC Fire Services from a ProbeMatrix four-level human-factors investigation.

Prepared for: ONGC Fire Services

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For installation Chief Fire Officers, Fire Officers and the Fire Services planning staff of Oil and Natural Gas Corporation Ltd.

Learning purposes: not a statutory document. Derived from a ProbeMatrix investigation export; learnings are evidence-linked but should be validated by the organisation before adoption.

Status and source. This is an independent learning analysis prepared by Eyekyam Risk Resolutions from publicly available information about the incident, analysed with ProbeMatrix, a human-reviewed method built on the four-level Human Factors Analysis and Classification System (HFACS). It is **not** ONGC's investigation and does not present ONGC's findings. Page references in the evidence come from the inquiry committee material that the analysis drew on. The ProbeMatrix investigation is marked *in progress*, so treat every learning here as a draft. We would welcome corrections from ONGC Fire Services.

Corrections to the source data. The export's summary sheet gives the date as 24 August 2020 and the incident type as "aviation". The narrative and the evidence show a hydrocarbon vapour release and fire on **24 September 2020**, and this document uses that date. One finding with a reviewer downgrade vote (FND-44219027, "not substantiated by evidence") is excluded.

People. The document names functions, not people or ranks. The people on duty that night were working within the plans, equipment and authority they were given. These learnings are about those systems.

Bottom line for Fire Services

1. **The fire-water system was healthy on the night, but only because the electric pumps ran.** Three of the four diesel fire-water pumps were under maintenance or unreliable, and one tripped after two minutes from a cause nobody had identified. If grid power had been lost, the backup was not there. *(Learning FS-1)*
2. **The drill records could not show whether the station was ready.** The SAP drill reports held dates but no observations or actions, and a parallel informal record did not follow the approved guidelines. *(FS-2)*
3. **The fire station was told at about 02:15, 53 minutes before ignition.** The available record shows one pre-ignition fire-service action: a halt on vehicle movement in the plant at about 02:35. A large unignited vapour cloud is a fire-service phase in its own right, and the plan should say so. *(FS-3)*
4. **Response times on the night cannot be measured reliably.** Control-room clocks disagreed by 2 to 12 minutes, the only camera was lost 7 seconds after ignition, and the record contains no fire-service log. You cannot improve a response you cannot time. *(FS-4)*
5. **The deepest causes sit at the interfaces.** The emergency plan had no isolation criteria, the isolation valves could only be worked by hand inside the cloud, gas detection only displayed an alarm, and warnings were relayed through intermediate points. Fire Services do not own these, but can influence each one through ERDMP review, drills and joint inspections. *(IF-1 to IF-6)*

Part A. The accident: what happened

A1. The plant and the area

The Hazira Gas Processing Complex receives gas and condensate from Mumbai Offshore through two submarine trunk lines: a 36-inch line commissioned in 1986 (about 230 km) and a 42-inch line commissioned in 1996 (about 244 km). Its capacity is about 46.9 million standard cubic metres a day, and it supplies the HVJ pipeline network, a fertiliser plant and a power station.

The **Gas Terminal (Phase I)** is the inlet section, where the offshore stream is received and separated. Downstream of the slug catcher, a **condensate metering skid** carries two parallel 6-inch meters, 310FT01 and 310FE2. Both are part of the original 1986 plant. The skid stands at ground level, north-west of the Gas Terminal control room.

The area was watched by a **single pan-tilt-zoom CCTV camera** that takes about 90 seconds to complete a turn, and by CISF perimeter security from watch towers and gates. Gas and flame detectors covered the area, but they showed alarms on a local annunciator panel only. They were not recorded in the control system and were not connected to any automatic shutdown or fire-suppression action. [EVD-0210; EVD-3452; EVD-8657]

A2. The week before

- **17 September 2020.** Operations raised a maintenance notification for a malfunction of meter 310FT01. [EVD-0591]
- **18 September 2020.** The meter's retaining-plate gasket was replaced. The original manufacturer's gasket specification for this 1986 rectangular-slot design was not held at site, so a locally bought O-ring was used, as had become routine practice. The system was then pressurised to operating pressure only. **No hydrostatic test to design pressure was carried out.** [EVD-8645; EVD-2894]
- **19 September 2020.** The meter returned to service and was in use when the release happened. [EVD-0185]
- This was the **third gasket replacement on this meter in about 22 months** (after November 2018 and February 2019). [EVD-1494]

None of this was visible to the fire station. It matters to Fire Services for one reason: the release came from a recently repaired joint on old equipment. That is a pattern a fire pre-plan can flag. *(See the self-check under FS-3.)*

A3. The night of 23–24 September

Before the leak. The night shift in the Gas Terminal control room took over at 20:00 with two engineers. At 02:11:43 the CCTV showed the facilities intact. [EVD-9587]

02:12:50: the leak begins. CCTV recorded the first visible vapour leaving the metering skid. No alarm had sounded yet. [EVD-1284]

02:15: people see it first. A CISF guard on the north-west watch tower heard a sharp hissing and saw a vapour cloud rising behind the Gas Terminal. The guard informed the CISF main gate, **the Fire Station** and the main LPG control room. At the same moment, a CISF guard at the Gas Terminal gate heard the noise, saw the cloud, and informed the CCTV control room and the CISF main gate. Neither report went directly to the Gas Terminal control room, which controlled the leaking equipment. [EVD-4286; EVD-6234]

Over the next minute the CCTV showed the cloud spreading from the skid to the area around the control room, onto the road and through the piping, then thickening. [EVD-9941]

02:18–02:20: the control room learns of it. The CISF officer in the CCTV control room confirmed heavy leakage on the monitor and telephoned the Gas Terminal control room at 02:18. At about 02:20 the gas detection annunciator alarmed in the control room. That was about 7 minutes after the leak began and 5 minutes after people first saw it. The two engineers heard the alarm and took the CCTV call at the same time. [EVD-8871; EVD-8657]

02:20–02:35: the cloud blocks the way. Both engineers went out to find the source. The cloud was so dense, and the smell so strong, that they could not get close, and visibility was very poor. They returned and telephoned the shift in-charges and the main control room. [EVD-1460; EVD-9220]

About 02:35: senior shift staff arrive and cannot get in either. The shift in-charge team reached the Gas Terminal gate but could not enter because of the gas. One of them **informed the Fire Station to stop all vehicle movement in the plant**. The overall shift in-charge told the control room to close the 36-inch pressure control valve and leave the building, informed the Fire Station, and began calling senior management in turn. [EVD-9612; EVD-7340]

02:40–03:00: escalation. CISF barricaded the road to the slug catcher area. Senior engineers and managers were called and arrived one after another. At about 03:00 a senior operations manager took over as Site Incident Controller. Throughout this period the emergency plan offered no criterion for when to isolate and named no one below the top of the chain with standing authority to do so. The isolation valves for this area could only be operated locally, at the valve. [EVD-4052; EVD-3932]

03:00:44. The CCTV showed vapour "all around even at height". [EVD-8213]

03:01: first isolation. The shift in-charge team closed **MOV 54**, the condensate line from the slug catcher to the metering skid. This was **46 minutes after the leak was first seen**. At 03:04 the 36-inch pressure control valve was closed from the control room, and offshore was asked to stop supply through both trunk lines. [EVD-9612; EVD-6414]

03:08:15: ignition. The cloud ignited, **4 minutes after both isolations were complete**. Isolating upstream did not remove the vapour already in the air. The probable ignition source was a flameproof light fitting near the skid. Three of its bolts were missing and its glass case stood open by 5 to 6 cm. Seven seconds later the CCTV cable burned through and the recording stopped. [EVD-4507; EVD-6916; EVD-2752]

A4. The fire and its control

Time	Event
03:08:15	Ignition. Major fire at the metering skid and surrounding piping.
~03:10	**Level-III (off-site) emergency** declared by the Site Incident Controller with senior plant management. Mutual aid called. Offshore told to stop supply in both trunk lines. Process trains, cogeneration turbines and boilers shut down; cooling towers kept running; grid power stayed available.
~03:15	The Chief Incident Controller (senior plant management) takes command in the Emergency Control Room. Isolation and depressurisation of connected units begins.
~03:30	Trunk-line sectionalising valves at Gavier (36") and Abhwa (42") confirmed closed by mobile teams. Depressurisation and condensate evacuation under way.
~03:50	**Active fire-cooling operations begin** (about 42 minutes after ignition).
~04:30	Phase-I to Phase-II condensate header interconnection closed.
~05:30	MOVs 50 and 52 closed: bulk hydrocarbons isolated from the Phase-I slug catcher.
~06:00	**Fire extinguished**; cooling continues.
~07:00	Main 36" inlet valve (MOV 03) closed once the area could be approached; zero LEL confirmed.
07:30	**All-clear** declared by the Chief Incident Controller after a complete head-count.

Source: the initial incident report in the investigation narrative, reconciled from the control-room log, DCS history, CCTV, CISF logs and statements.

Fire water. Fire-water tank levels and header pressure stayed in their normal bands throughout. Three electric fire-water pumps (47PE003 A, C and D) carried the cooling load. Of the four 410 m³/h diesel fire-water pumps, two were under maintenance, one ran for 30 minutes, and the fourth trips after two minutes for a reason that had not been identified or recorded. [EVD-2432]

Casualties. None among plant personnel or on-shift contractors. CISF confirmed no injuries inside the plant boundary. At the time of the initial report, the status of the settlement outside the perimeter was still being checked with the district authorities.

Damage. The metering skid area was a total loss (meters, instruments, immediate piping and cabling). Electrical fittings within about 15 m, including the CCTV camera and cable, were significantly damaged. The 36" and 42" trunk lines, the slug catcher body, the control rooms and the Emergency Control Room were undamaged. [EVD-1394; EVD-5780]

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A5. The night in four numbers

Interval	Duration
Leak onset (02:12:50) to first human detection (02:15)	~2 min
First detection (02:15) to first isolation (03:01)	**46 min**
Fire Station first informed (~02:15) to ignition (03:08:15)	**~53 min**
Ignition (03:08:15) to active cooling (~03:50)	**~42 min**
Ignition to fire extinguished (~06:00)	~2 h 52 min

A6. What the record does not tell us

This is a desk analysis. The available record does **not** describe:

- what the Fire Station did between receiving the 02:15 call and the start of active cooling at about 03:50: turnout, where crews and tenders were staged, and any water-curtain or dispersion decision;
- whether the fire service was consulted on isolation or ignition control before 03:08;
- why active cooling began about 42 minutes after ignition, which may reflect approach conditions, staging, the order of isolation, or simply how the time was recorded;
- when mutual aid arrived and what it did;
- whether the Fire Station receives gas-detection alarms directly.

These gaps are not criticisms. The fire was contained with no injuries inside the plant. They are the questions a fire-service debrief would answer, and several learnings below depend on them.

Part B. Fire-led learnings (owned by Fire Services)

Each learning gives the evidence, why things made sense at the time, what good looks like, recommendations rated on the hierarchy of controls, a leading indicator, and a self-check for any ONGC installation.

FS-1. The diesel fire-water backup was mostly unavailable, and nobody had diagnosed why

What happened. On the night, three of the four 410 m³/h diesel fire-water pumps were unavailable or unreliable: two under maintenance, and one tripping after two minutes from an unidentified cause, with no corrective action recorded. Only one diesel pump was in a usable state, and it ran for 30 minutes. Electric pumps carried the whole cooling load. Because grid power stayed on, the gap did not affect the outcome. **Evidence strength: direct (committee record).** [EVD-2432; FND-B955FBD6; FND-E7FB469A; FND-D9E0CE71; FND-DF7337F2; FND-4317808B; FND-D8BC7096; and 7 related instances]

Why it made sense at the time. Fire-water tank level and header pressure looked healthy day to day, because the electric pumps always ran. A diesel pump that trips after two minutes looks fine on a short weekly run. Without a rule that a degraded diesel pump is an impairment, it stays a routine maintenance job.

Why it matters to Fire Services. Diesel pumps exist for the night the grid fails, and a major fire near electrical equipment is exactly that night. At Hazira the fire damaged electrical fittings within about 15 m. A larger or differently placed fire could have taken out power to the electric pumps.

What good looks like. Diesel fire-water capacity is treated as a safety-critical barrier. Each pump's availability is known at every shift handover. A trip or failure is diagnosed and its cause recorded. When standby capacity falls below the design basis, an impairment is declared and compensating measures are put in place (for example, mutual-aid water arranged, a portable pump staged, high-risk work suspended).

Recommendations

#	What should change	Barrier	Tier
FS-1a	Restore the diesel fire-water pumps to design availability, and replace units that cannot be made reliable, so standby capacity does not depend on grid power	Fire-water supply (mitigation)	Engineering
FS-1b	Make "diesel fire-water capacity below design basis" a declared impairment with defined compensating measures and an escalation route	Fire-water supply	Administrative
FS-1c	Require every fire-pump trip or failed start to have a recorded cause and corrective action before it is closed	Fire-water supply	Administrative
FS-1d	Test diesel pumps on full-duration runs at rated flow, not start-only checks, so a 2-minute trip shows up	Fire-water supply	Administrative

Leading indicator. Percentage of hours in the month when installed diesel fire-water capacity meets the design basis. Number of pump trips or failed starts still without a recorded cause after 7 days.

Check at your installation. How many diesel fire pumps can deliver rated flow right now, for the full test duration? If the grid failed in the next hour, how long could you cool a major fire? Is a degraded diesel

pump an impairment in your system, or a maintenance job?

See also: OISD-STD-116, *Fire Protection Facilities for Petroleum Refineries and Oil/Gas Processing Plants*, on fire-water pump capacity, standby and testing. Check against the current edition.

FS-2. Drill records did not record what drills found, so readiness could not be shown or improved

What happened. Monthly fire drill reports existed in SAP, but held no data on observations, actions or recommendations. A separate dataset was kept that did not follow the approved guidelines. Neither record could show whether drills had ever exercised a large hydrocarbon release, how long key steps took, or whether problems found in drills were fixed. **Evidence strength: direct.** [EVD-2130; FND-332DC593; FND-A532B8F8; FND-DC0751CF; FND-BFABC63C; FND-C69E0C67; FND-B8B75D91; FND-BC32F8F7; FND-7F27ED51; FND-88D0E4B1]

Why it made sense at the time. Drills were being held. A system that asks only for a date and a tick makes "drill done" the measure, and the useful notes end up in a separate, informal place.

Why it matters to Fire Services. The night's critical weaknesses all sat in the pre-ignition phase: the slow relay of the alarm, no isolation criteria, valves that could only be reached on foot, and the cloud blocking access. These are the kind of things a well-designed drill finds and a well-kept record carries forward. Without the record, the organisation could not tell whether its drills had tested them.

What good looks like. One drill record per drill that captures the scenario, timed decision points, observations, actions with owners, and closure. Actions from drills are tracked like any other safety action. Scenarios rotate through the installation's credible major events, including an **unignited** release.

Recommendations

#	What should change	Barrier	Tier
FS-2a	Replace the date-only SAP drill record with a structured record: scenario, timed decision points, observations, actions, owner, closure date	Emergency readiness assurance	Administrative
FS-2b	Retire the parallel informal dataset, so there is one record that conforms to the approved guidelines	Emergency readiness assurance	Administrative
FS-2c	Set a drill programme that covers each credible major-event scenario in the ERDMP, including large unignited vapour releases at night (see the drill pack)	Emergency response	Administrative

No engineering-tier control fits here: this is about learning from drills. FS-1a, IF-2 and IF-3 carry the stronger controls.

Leading indicator. Percentage of drills in the quarter with timed decision points recorded. Percentage of drill actions closed within 30 days.

Check at your installation. Pick your last three drill records. Could someone who wasn't there say what was learned and whether it was fixed? When did you last drill an unignited release, rather than a fire?

See also: PNGRB *Codes of Practices for Emergency Response and Disaster Management Plan (ERDMP) Regulations, 2010* (as amended), on drills and mock exercises. Check the current consolidated text and its applicability to your installation.

FS-3. The 53 minutes before ignition were a fire-service phase, but the plan did not treat them as one

What happened. The Fire Station was among the first to be told, by the watch-tower guard at about 02:15 [EVD-4286; FND-5AB5DFA3; FND-95F9205E]. At about 02:35 the shift in-charge team told the Fire Station to stop all vehicle movement in the plant [EVD-9612; FND-305216AF]. That was a sound ignition-control decision. By 03:00 vapour filled the area "even at height" [EVD-8213]. The cloud ignited at 03:08:15, four minutes after the upstream isolations, most probably on a damaged flameproof light fitting [EVD-4507; EVD-6916]. The record describes no other fire-service activity before ignition (see A6). **Evidence strength: direct for the facts; the learning is inferred, and the [GAP] in A6 applies.**

Why it made sense at the time. Most fire-service plans and drills begin at "fire". A vapour cloud with no flame reads as an operations problem: find the leak and isolate it. In that frame the fire service stands by. The emergency plan named the major-leak scenario but set out no actions for it. [EVD-4052]

Why it matters to Fire Services. Between release and ignition, the fire service holds the levers that most affect the outcome:

- **ignition-source control:** vehicles, hot work, and the question of non-flameproof electrics, which needs an advance decision agreed with electrical engineers, because switching under a cloud is itself a hazard;
- **staging:** upwind, outside the expected cloud envelope, with turnout routes that do not cross it;
- **dispersion:** a decision on water curtains or monitors to dilute or deflect the cloud away from known ignition sources;
- **exclusion and accounting:** who is inside the cloud area, and a cordon;
- **asking for isolation status:** what has been closed, what can still be closed, and from where.

At Hazira, ignition came after isolation. Once a cloud has formed, closing valves upstream does not make it safe. That is when fire-service measures matter most.

What good looks like. The fire pre-plan and the ERDMP treat "large unignited release" as its own scenario, with fire-service actions, triggers and timings. The Fire Station's first action on a gas-release call is a scripted set of questions and moves, not waiting.

Recommendations

#	What should change	Barrier	Tier
FS-3a	Write a fire-service procedure for large unignited hydrocarbon releases: ignition-control checklist, staging rules, water-curtain decision criteria, cordon and accounting, and the questions to ask the control room	Ignition prevention; escalation control	Administrative
FS-3b	Pre-plan fixed or portable water curtains/monitors for high-inventory skids such as metering, slug catcher and pig receivers, so dispersion is available without approaching the cloud	Ignition prevention (dispersion)	Engineering
FS-3c	Agree in advance with electrical engineering which area lighting and power circuits may be isolated during a vapour release, and from where (outside the hazardous area)	Ignition prevention	Administrative
FS-3d	Exercise the procedure in a night drill (drill pack, Drill 1)	Emergency response	Training

Leading indicator. In drills, the time from gas-release call to completion of the ignition-control checklist, and the time to a recorded water-curtain decision.

Check at your installation. If a cloud formed at your highest-inventory skid tonight, which fire-service actions are written down for the time before ignition? Where would crews stage, and would the route cross the cloud? Is there a water curtain you could use without going in?

FS-4. The night's response cannot be timed reliably, so it cannot be improved by measurement

What happened. DCS clocks in the different control rooms disagreed by 2 to 12 minutes, and "data matching is difficult" [EVD-5532; FND-0C37BB1E; FND-EE34220F; FND-2B215AA5; FND-D9A35E88; FND-28F6EE8A; FND-1A5ECA30]. The single CCTV camera recorded until 03:08:22 and was then lost with its cable [EVD-2752; FND-AA71AC71]. The timeline after ignition comes from logs and later accounts, with approximate times ("~03:50", "~06:00"), and the record contains no fire-service log. **Evidence strength: direct for clocks and CCTV; the absence of a fire-service log is a [GAP].**

Why it made sense at the time. In a real fire, crews fight the fire, and time-stamped logging falls to whoever has a free hand. Clock drift between control systems is invisible until someone tries to reconstruct an event.

Why it matters to Fire Services. Every time standard in the drill pack (alarm to turnout, ignition to cooling) needs a reliable clock. Without one, a 42-minute ignition-to-cooling interval cannot be explained, defended or improved.

What good looks like. The fire control room keeps a time-stamped incident log, started on the first call, using a clock synchronised with the plant's systems. In a major incident a named role keeps the log. Drill

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timings use the same clock.

Recommendations

#	What should change	Barrier	Tier
FS-4a	Start a time-stamped fire-service incident log on the first call, with a dedicated loggist in major emergencies	Emergency response; learning	Administrative
FS-4b	Synchronise fire-control-room clocks with plant DCS and CCTV time (for example to a common network time source), and check the drift monthly	Emergency response; learning	Engineering (interface with instrumentation)

Leading indicator. Maximum clock drift found at the monthly check. Percentage of incidents and drills with a complete time-stamped fire-service log.

Check at your installation. Compare the fire-control-room clock with the DCS clock today. What is the difference? After your last incident, could you show when the first water went on?

Part C. Interface learnings (owned by others, influenced by Fire Services)

These learnings have the strongest evidence in the investigation, but Fire Services do not own them. Each states **who owns it** and **how Fire Services can influence it**, typically through ERDMP review, drills, joint inspection and the fire-safety audit. The pack makes no recommendation that Fire Services change anything they do not control.

IF-1. The emergency plan named the major-leak scenario but gave no isolation criteria and no delegated authority

Owner: plant management, as owner of the DMP/ERDMP.

What happened. The on-site Disaster Management Plan included hydrocarbon-leak guidance but did "not include isolation/stoppage of facilities/pipelines". The ERDMP's Level-3 scenarios included heavy gas leaks without a decision on isolation. No role below the top of the chain had standing authority to isolate, so the decision went up by telephone, one call at a time. **Evidence strength: corroborated.** [EVD-4052; EVD-3932; FND-DF1144C3; FND-82761CC5; FND-730D0DCA; FND-6472CD15; FND-E009D26B; FND-5C7B163D; and 11 related instances]

Fire Services' lever. As ERDMP stakeholders, ask for each major-release scenario to state: the isolation trigger, who may isolate without seeking permission, and which fire-service actions run in parallel. Then test it in a drill (Drill 1 and Drill 4).

Check at your installation. Read your ERDMP's major-release scenario. Does it say who closes which valve, on what trigger, without phoning anyone?

IF-2. The isolation valves could only be closed by hand, inside the area the cloud made unreachable

Owner: operations and engineering.

What happened. The MOVs at the Gas Terminal "are not having remote operation from control room during emergency". The engineers and then the shift in-charge team were turned back by the cloud, and MOV 54 was closed at 03:01. **Evidence strength: corroborated.** [EVD-3932; EVD-9612; FND-31405D96; FND-044D1DEC; FND-6DB250E6; FND-02310148; FND-DE978BF5]

Fire Services' lever. Fire pre-plans should list, for each high-inventory section, which isolation points can be operated from outside the likely cloud. Where none exist, record that as a fire-safety observation. Drills should reveal it (Drill 1).

Check at your installation. For your three highest-inventory sections, can isolation be done from a safe location? Has a drill tested that?

IF-3. Gas detection only displayed an alarm, and it alarmed after people had already seen the cloud

Owner: instrumentation / F&G engineering, with operations.

What happened. Hydrocarbon and H2S detectors showed on a local annunciator panel and were not recorded in the DCS. The F&G system had no provision for automatic plant shutdown or for activating fire suppression. The first alarm came at about 02:20, about 7 minutes after the leak began and 5 minutes after the watch-tower report. The inquiry committee recommended interlocking F&G detection to automatic shutdown and suppression (recommendation A.1, p.34). **Evidence strength: corroborated.** [EVD-3452; EVD-8657; EVD-1284; FND-6155A1C2; FND-C5000A82; FND-3D785319; FND-9399FA02; FND-B5C0CE8F; FND-119E6CF7; and 14 related instances]

Fire Services' lever. Ask whether gas and fire alarms from high-hazard areas are repeated at the fire control room. The record does not show whether Hazira's were [GAP]. Support the interlock recommendation where fire suppression is involved, and include F&G-to-suppression function tests in the fire-protection test programme.

Check at your installation. Does your fire control room see a gas alarm at the same moment the process control room does? Does any detector trigger suppression automatically?

IF-4. People saw the release first, but their warnings reached the control room late and through intermediate points

Owner: CISF / security, with operations.

What happened. Two CISF posts saw and heard the release at 02:15. One informed the main gate, the Fire Station and the LPG control room. The other informed the CCTV control room and the main gate. The

Gas Terminal control room heard at 02:18 to 02:20, after the CCTV control room had verified the report on its monitor. **Evidence strength: corroborated.** [EVD-4286; EVD-6234; EVD-8871; FND-BC00FF3A; FND-5AB5DFA3; FND-95F9205E; FND-2F736E5F; FND-439E0DF7; FND-0A64F629]

Fire Services' lever. The Fire Station was told within minutes. That is good. Ask for a "one call reaches all" arrangement (a conference line, a broadcast, or a fixed call list) so that a security report of gas reaches the fire station, the controlling control room and the shift in-charge together. Rehearse it in the tabletop (Drill 4).

Check at your installation. If a watch tower sees a cloud, who hears first, and how many calls does it take for the control room that owns the equipment to know?

IF-5. One panning camera was the only view of the skid, and it was lost seconds after ignition

Owner: security and instrumentation.

What happened. A single PTZ camera with a 90-second pan covered the metering skid and control-room area. The CCTV cable burned through 7 seconds after ignition. The committee recommended CCTV for each plant, displayed in plant control rooms and at master control / the CISF station (p.36). **Evidence strength: corroborated.** [EVD-0210; EVD-2752; FND-38F0B929; FND-B59A077E; FND-FBC20DA0; FND-2E2F2A38; FND-A528DBE5; FND-D515D5E4; and 4 related instances]

Fire Services' lever. A fire officer directing the response from outside the cloud needs eyes on it. Ask for fire-control-room or ECR access to CCTV feeds, and for fire-rated routing of camera cables in high-hazard areas.

Check at your installation. Can your fire control room see the high-hazard areas live? Would the view survive the first minute of a fire?

IF-6. A damaged flameproof light fitting stayed in service in a hazardous area

Owner: electrical maintenance.

What happened. The probable ignition source was a flameproof HPMV light fitting with three bolts missing and its glass case open by 5 to 6 cm. The nearby fittings were intact. **Evidence strength: direct (physical evidence; ignition source "probable").** [EVD-6916; EVD-4507; FND-C7F16268; FND-74A80D24; FND-784A23E4; FND-ABF11182; FND-5E08D0F8; and 22 related instances]

Fire Services' lever. Fire-safety inspections and walk-downs pass these fittings every day. Agree with electrical maintenance that a visibly open or bolt-missing flameproof enclosure in a hazardous area is reported as a fire-safety defect, with a defined response time.

Check at your installation. On your next walk-down of a classified area, look at the flameproof fittings. Who would you tell if one were open, and how fast would it be fixed?

Part D. Questions Fire Services may wish to raise internally

1. What did our crews do between the first call and the start of cooling, and what would we do differently? *(A6)*
2. Is diesel fire-water capacity below design an impairment in our system today? *(FS-1)*
3. Do our drill records show what we learned, not just that we drilled? *(FS-2)*
4. Do our plan and drills cover the unignited cloud, not just the fire? *(FS-3)*
5. Who in our ERDMP may isolate without asking, and has a drill tested it? *(IF-1)*

Standards to check against

These are pointers only. The analysis does not test compliance with any clause.

- **OISD-STD-116**, *Fire Protection Facilities for Petroleum Refineries and Oil/Gas Processing Plants*: fire-water pumps, standby capacity, testing. Title confirmed from public listings; check the current edition.
- **PNGRB Codes of Practices for ERDMP Regulations, 2010**, as amended (2015, 2020; consolidated text 2025): emergency plan content, scenarios, drills. Confirm applicability to your installation.

Evidence trail and limits

- **Source:** ProbeMatrix investigation export `Hazira2020-findings-export-INC-35-2026-10-15.xlsx` (436 findings, 46 evidence items). Learnings rest on the finding (FND) and evidence (EVD) IDs cited. Several findings describe one condition, so counts are instance-level support, not separate problems.
- **Related register:** this pack is consistent with the full Learnings Register for the same investigation: FS-1 = LR-23, FS-2 = LR-24, IF-1 = LR-07, IF-2 = LR-09, IF-3 = LR-10, IF-4 = LR-13, IF-5 = LR-14, FS-4 draws on LR-15, IF-6 = LR-11.
- **Limits:** a desk analysis of documents. It does not replace site access, interviews or physical examination, and it cannot see the fire service's own records.

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